

Marwa Jalili, MPH

www.linkedin.com/in/marwa-jalili | San Diego, CA | marwa.jalilio1@gmail.com

Professional Experience

Nucleix

San Diego, CA

Assay Development Research Associate

January 2024 - Present

- Scaled qPCR assay from R&D to large-scale algorithm training and validation studies through cross-functional collaboration.
- Implemented stage-specific workflow controls to detect assay noise and rapidly flag study failures.
- Enhanced assay reproducibility by standardizing material specifications and QC measures.
- Analyzed large datasets using JMP and Python, generating insights to guide assay optimization.

Clinical Laboratory Technician

August 2022 - January 2024

- Designed QC specifications and functional assays for cfDNA extraction, purification, and qPCR reagents.
- Established acceptance criteria and data analysis pipelines for reagent qualification.
- Manufactured multiplex qPCR master mixes and DNA controls under aseptic conditions.
- Processed clinical samples for plasma separation and cfDNA extraction in CLIA-certified workflows.
- Authored SOPs and ensured compliance with CAP and CLIA inspections.

Scripps Research Translational Institute

San Diego, CA

Research Intern

June 2020 – January 2021

- Conducted systematic literature review on wearable technology for infectious disease detection.
- Identified health disparities in early diagnosis and prognosis of COVID-19 and other infections.
- Findings published in *The Lancet Digital Health*; presented research to scientific mentors.

Palomar Health

Escondido, CA

Pathmaker Intern

September 2020 – August 2022

- Assisted nurses, physicians, and technicians with patient care across multiple hospital departments (ICU, Neurology, Cardiology, ED).

Education

- San Diego State University, San Diego, CA.
 - Masters of Public Health, Epidemiology
- California State University San Marcos, San Marcos, CA.
 - B.S. Biological Sciences - Physiology, Minor: Chemistry

Graduation Date: May 2025

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Publications

- Radin, J. M., Quer, G., Jalili, M., Hamideh, D., & Steinhubl, S. R. (2021). The hopes and hazards of using personal health technologies in the diagnosis and prognosis of infections. *The Lancet Digital Health*, 3(7).
[https://doi.org/10.1016/s2589-7500\(21\)00064-9](https://doi.org/10.1016/s2589-7500(21)00064-9)

Skills

- **Epidemiology & Data Analysis:** Python, JMP, study design, statistical analysis
- **Public Health:** Health disparities, quality assurance, clinical study workflows
- **Molecular Biology:** qPCR, cfDNA extraction, biomarker development, SOP creation, NGS workflows
- **Instrumentation:** QuantStudio7, Nanodrop, Qubit, Fragment Analyzer, ABI7500
- **Compliance:** CLIA, CAP, OSHA, HIPAA